

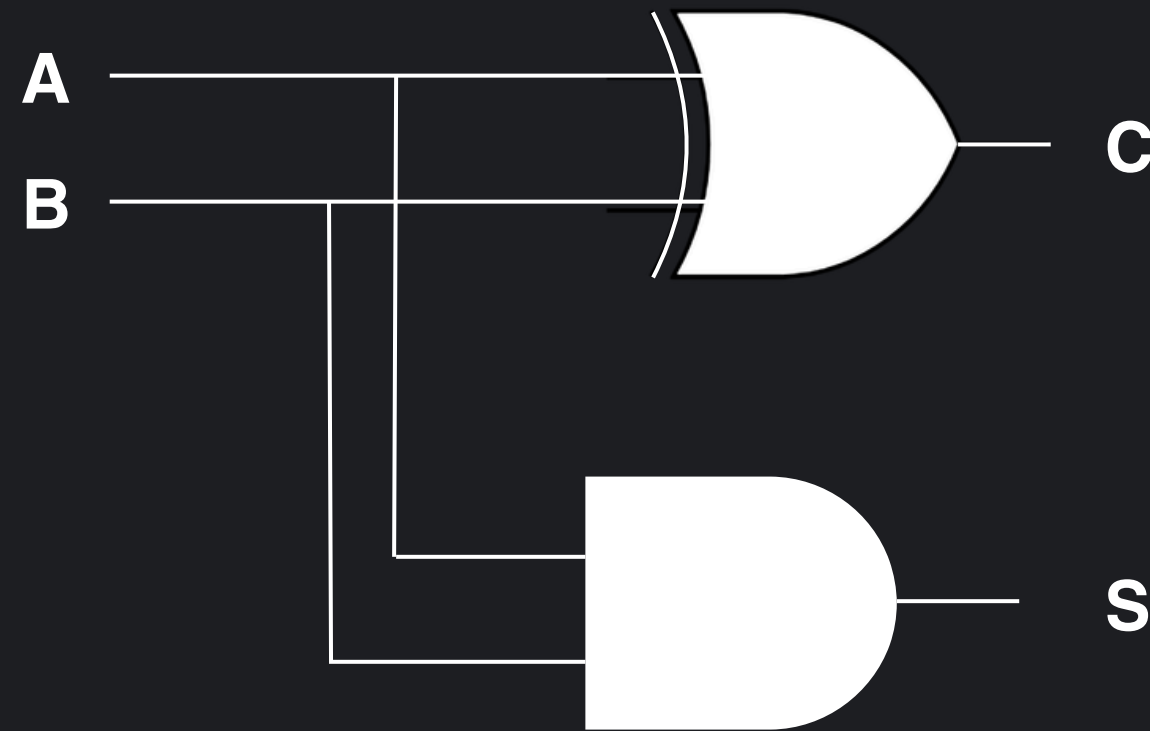
Half Adder



What is a Half Adder?

A Half Adder is a basic combinational logic element that performs the arithmetic addition of two binary bits. It produces two outputs, Sum and Carry, based on its two inputs which consist of the augend and the addend.

Truth Table



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Function of the Half Adder

A Half Adder performs a logical arithmetic addition on its two binary inputs. The main function of a Half Adder is to serve as a basic arithmetic element and produce an output signal (Sum and Carry) according to this rule:

Output Sum = 1 when the total of inputs (A, B) is 1.

Output Sum = 0 when the total of inputs is 0 or 2.

Output Carry = 1 when the sum of the inputs (A + B) is 2 (meaning both A and B are 1).

Output Carry = 0 when the sum of the inputs is less than 2.